

Q1

If the shortest distance between the lines $\mathbf{r} = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) + t(-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}})$ and

$\mathbf{r} = (\hat{\mathbf{i}} - 7\hat{\mathbf{j}} - 2\hat{\mathbf{k}}) + s(\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$ is equivalent to projection of $\mathbf{P} = -2\hat{\mathbf{i}} + 11\hat{\mathbf{j}}$ on $\mathbf{Q}$ then a possible vector $\mathbf{Q}$ is

- (1) $\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ (2) $5\hat{\mathbf{i}} - \hat{\mathbf{j}} + 3\hat{\mathbf{k}}$
 (3) $-1\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ (4) $3\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - \hat{\mathbf{k}}$

Q2

If R is a relation on the set of natural number such that $aRb \Leftrightarrow a = 3^k \cdot b$ for some integer k , then R is :-

- (1) Symmetric, transitive but not reflexive
 (2) Reflexive, symmetric but not transitive
 (3) Reflexive, transitive but not symmetric
 (4) An equivalence relation

Q3

Let a , b and c satisfy the system of equations $a + 2b + 3c = 6$, $4a + 5b + 6c = 12$ and $6a + 9b = 4$. If the roots of the equation $(a + b + c)x^2 - abcx + (a^{-1} + b^{-1} + c^{-1}) = 0$ are α and β , then $\frac{1}{\alpha} + \frac{1}{\beta}$ is equal to

- (1) 243 (2) 100 (3) $\frac{243}{12}$ (4) $\frac{100}{243}$

Q4

Two circles are given as $x^2 + y^2 + 14x - 6y + 40 = 0$ and $x^2 + y^2 - 2x + 6y + 7 = 0$ with their centres as C_1 and C_2 . If equation of another circle whose centre C_3 lies on the line $3x + 4y - 16 = 0$ and touches the circle C_1 externally and also $C_1C_2 + C_2C_3 + C_3C_1$ is minimum, is $x^2 + y^2 + ax + by + c = 0$ then the value of $(a + b + c)$ is

- (1) 2 (2) 0 (3) 16 (4) None of these

Q5

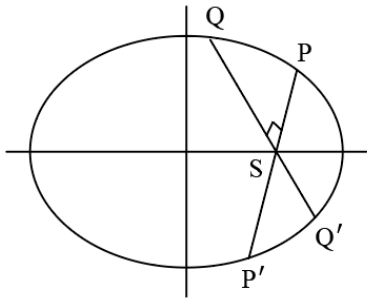
If A_1, A_2, A_3, A_4 are any four consecutive binomial coefficients in the expansion of $(1 + x)^n$, then

$$\frac{A_1}{A_1 + A_2} + \frac{A_3}{A_3 + A_4} - \frac{2A_2}{A_2 + A_3} =$$

- (1) $n - 2$ (2) 1 (3) -1 (4) 0

Q6

In an ellipse $\frac{x^2}{10} + \frac{y^2}{6} = 1$. If focal chords PSP' and QSQ' are at right angles to each other, then $\frac{1-e^2}{(SP)(SP')} + \frac{1-e^2}{(SQ)(SQ')} = \frac{N}{15}$. The value of [N] is (where [.] is greatest integer function)



- (1) 2 (2) 4 (3) 3 (4) 1

Q7

The positive real number which satisfy the equation $\frac{1}{x^2+4x-50} + \frac{1}{x^2+4x-66} = \frac{2}{x^2+4x-90}$

- (1) 5 (2) 8 (3) 6 (4) 4

Q8

Let A(2, 3, -1) be a point in space and B be a point on the line $\frac{x-4}{3} = \frac{y-2}{4} = \frac{z-4}{-5} = t$. The vector $\vec{AB}$ is perpendicular to the line $\frac{x-3}{3} = \frac{y-8}{1} = \frac{z-3}{1}$. Evaluate $|4t|$.

- (1) 4 (2) 5 (3) 7 (4) 1

Q9

If O, G, S are respectively the orthocentre, centroid and circumcentre of a triangle whose vertices are A(2, 3), B(2, 4) and C(4, 3), then $AO^2 + 9BG^2 + 4CS^2 =$

- (1) 11 (2) 10 (3) 13 (4) 12

Q10

If z is the complex number satisfying $|z - 3 - 4i| \leq 10$ and $\alpha = \sin^{-1}(\sin(|z|_{\max}))$, $\beta = \cos^{-1}\left(\cos\left(-\frac{|z|_{\max} + |z|_{\min}}{3}\right)\right)$, then -

- (1) $\alpha - 3\beta = \pi$ (2) $\alpha - 3\beta = -\pi$
(3) $\sin^2 \alpha + \cos^2 3\beta = 2$ (4) $\sin^2 \alpha + \cos^2 3\beta = 1/2$

Q11

While calculating the mean of a set of 15 observations, the mean was obtained as 9 and the sum of squares of the observations was found to be 1347. Later, it was found that two observations were wrongly taken as 4 and 6 instead of the correct values 10 and 15. Find the variance of the corrected data.

- (1) 6 (2) 8 (3) 10 (4) 12

Q12

The area of the figure bounded by the parabolas $x = -2y^2$ and $x = 1 - 3y^2$ is

- (1) $\frac{4}{3}$ sq units (2) $\frac{2}{3}$ sq units (3) $\frac{3}{7}$ sq units (4) $\frac{6}{7}$ sq units

Q13

For $a, b > 0$, let $5a - b, 2a + b, a + 2b$ be in A.P. and $(b + 1)^2, ab + 1, (a - 1)^2$ are in G.P., then compute $(a^{-1} + b^{-1})$

- (1) 7 (2) 6 (3) 5 (4) 2

Q14

2 players A, B tosses a fair coin in cyclic order $A, A, B, A, A, B, \dots$ till a head appears. If the probability that A gets head first is p , then $\frac{24}{p}$ is equal to

- (1) 28 (2) 18 (3) 27 (4) 14

Q15

Determine the value of the expression $\frac{12}{\sqrt{3}\pi^2} \int_0^{\frac{\pi}{2}} \frac{x dx}{\sin^4 x + \cos^4 x + \sin^2 x \cos^2 x}$.

- (1) 1 (2) 2 (3) 3 (4) 4

Q16

Let $f(x)$ be a polynomial one-one function such that

$$f(x)f(y) + 2 = f(x) + f(y) + f(xy), \forall x, y \in \mathbb{R} - \{0\}, f(1) \neq 1, f'(1) = 3$$

Let $g(x) = \frac{x}{4}(f(x) + 3) - \int_0^x f(x)dx$, then

- (1) $g(x) = 0$ has exactly one root for $x \in (0, 1)$
 (2) $g(x) = 0$ has exactly two root for $x \in (0, 1)$
 (3) $g(x) \neq 0, \forall x \in \mathbb{R} - \{0\}$
 (4) $g(x) = 0, \forall x \in \mathbb{R} - \{0\}$

Q17

A function $f(x)$ defined on $(-\frac{\pi}{2}, \frac{\pi}{2})$ with real values and has a primitive $F(x)$ such that

$$f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1 + \sin x)^2}, \text{ then } f(x) \text{ is}$$

- (1) $\frac{2 \cos x}{(1 + \sin x)^2} + ce^{-\sin x} \cos x$ (2) $\frac{2 \cos x}{(1 - \sin x)^2} + ce^{-\sin x} \cos x$
 (3) $\frac{-2 \cos x}{(1 + \sin x)^2} - ce^{-\sin x} \cos x$ (4) $\frac{-2 \cos x}{(1 - \sin x)^2} - ce^{\sin x} \cos x$

Q18

Determine the value of the limit: $\lim_{x \rightarrow 0} \frac{1}{\sqrt{2}} \operatorname{cosec} x \left(\sqrt{1 + 2 \sin x + \cos x} - \sqrt{1 - 2 \sin x + \cos x} \right)$.

- (1) 0 (2) 1 (3) $\sqrt{2}$ (4) 2

Q19

The area bounded by curves $y = \left[6 + x \left[\frac{1}{x}\right]\right]$, $y^2 - 18x + 18 = 0$ and $6x - 5y - 6 = 0$, (where $[\cdot]$ denotes the greatest integer function) is A , then $A =$

- (1) 12 (2) 13 (3) 9 (4) 11

Q20

$\vec{a}$, $\vec{b}$ and $\vec{c}$ are three vectors with magnitude $|\vec{a}| = 4$, $|\vec{b}| = 4$, $|\vec{c}| = 2$ and such that $\vec{a}$ is perpendicular to $(\vec{b} + \vec{c})$, $\vec{b}$ is perpendicular to $(\vec{c} + \vec{a})$ and $\vec{c}$ is perpendicular to $(\vec{a} + \vec{b})$. It follows that $|\vec{a} + \vec{b} + \vec{c}|$ is equal to:

- (1) 9 (2) 6 (3) 5 (4) 4

Q21

Let $A = \{1, 2, 3, 4, 5, 6\}$ number of functions f from A to A such that $f(m) + f(n) = 7$, whenever $m + n = 7$ is

Q22

Number of roots of the function $f(x) = \frac{1}{(x+1)^3} - 3x + \sin x$ is

Q23

Let P be a 3×3 real matrix such that $V^T P V = 0$ for every 3×1 column vector V . If $P \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ -7 \end{pmatrix}$ and

$P \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -3 \end{pmatrix}$, and it is given that $\det(\text{adj}(2(P + I)))$ can be written in the form $2^\alpha 3^\beta 5^\gamma$, where α, β, γ are

natural numbers, determine the value of $\alpha^2 + \beta^2 + \gamma^2$.

Q24

Let f be a differentiable function for $x \geq 0$ such that $3(x+1)^2 f(x) - (x+1)^2 = 8 \int_0^x (t+1) f(t) dt$. Find the value of $f(26)$.

Q25

If p, q, r are positive numbers satisfying $pqr = 3^{-5/2}$, then minimum value of $p^2 + 9q^2 + 6pq + 2r^2$ is

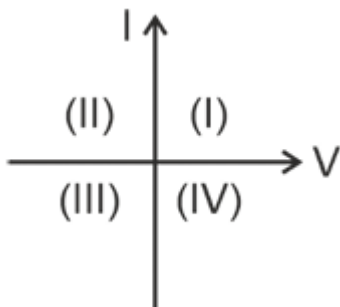
Q26

An electron in a hydrogen-like atom transitions between three distinct energy levels, P, Q, and R, where $E_P > E_Q > E_R$. When the electron moves from state P to state R, it emits a photon with a wavelength of $3000\text{\AA}$. If the transition from state Q to state R results in the emission of a photon with a wavelength of $9000\text{\AA}$, what is the wavelength of the photon emitted when the electron transitions from state P to state Q?

- (1) $3000\text{\AA}$ (2) $4500\text{\AA}$
(3) $5500\text{\AA}$ (4) $2000\text{\AA}$

Q27

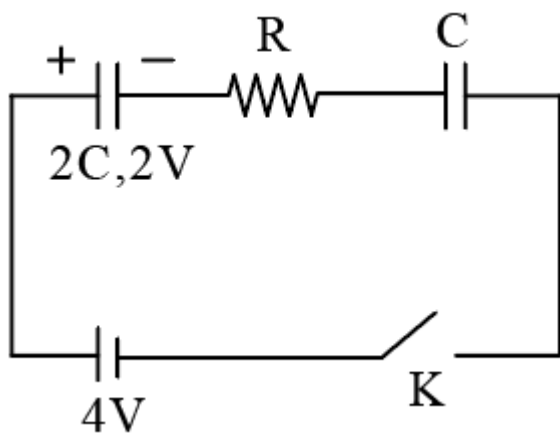
Consider the following statements A and B and identify the correct answer:



- A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.
B. In a reverse biased pn junction diode, the current measured in (μA), is due to majority charge carriers.
- (1) A is incorrect but B is correct (2) Both A and B are correct
(3) Both A and B are incorrect (4) A is correct but B is incorrect

Q28

A capacitor of capacitance $2C$ is charged to a potential 2 V and then connected to an uncharged capacitor of capacitance C with a resistance R to a battery of emf 4 V as shown in adjoining figure. The key is closed at $t = 0$. The charge on uncharged capacitor as a function of time will be



- (1) $4CV \left(1 - e^{-\frac{t}{2RC}}\right)$ (2) $\frac{4CV}{3} \left(1 - e^{-\frac{t}{2AC}}\right)$
(3) $4CV \left(1 - e^{-\frac{3t}{2RC}}\right)$ (4) $\frac{4CV}{3} \left(1 - e^{-\frac{3t}{2RC}}\right)$

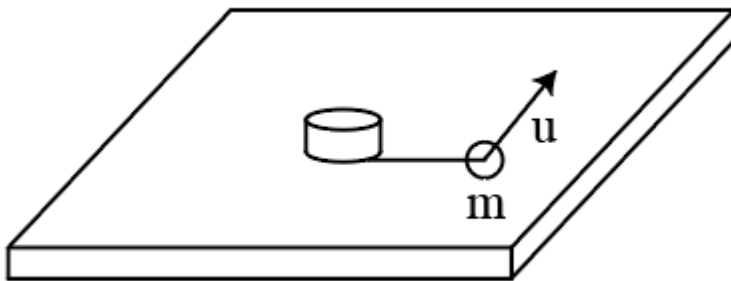
Q29

An AC source with angular frequency ω is connected to a series LR circuit. The power factor of the circuit is $\frac{1}{\sqrt{10}}$. If the angular frequency of the source is doubled to 2ω , what will be the new power factor of the circuit?

- (1) $\frac{1}{\sqrt{10}}$ (2) $\frac{1}{\sqrt{37}}$ (3) $\frac{1}{\sqrt{13}}$ (4) $\frac{1}{\sqrt{26}}$

Q30

A particle of mass m is attached to a long ideal string. The other end of the string is joined to a vertical cylinder fixed on a frictionless horizontal plane as shown in the figure. When the particle is projected with speed u in a direction perpendicular to the string, the particle continues to move on a spiral path of decreasing radius till it strikes the cylinder. While the particle is moving on the spiral path, how will the tension in the string and speed of the particle change?



- (1) Both will remain constant
 (2) Both will increase
 (3) Tension will increase and speed will decrease
 (4) Tension will increase and speed will remain constant

Q31

A system of binary stars of masses M_1 and M_2 are moving in circular orbits of radii r_1 and r_2 respectively. If T_1 and T_2 are the time periods of revolution of masses M_1 and M_2 respectively and if $\frac{r_1}{r_2} = 2$, then find $\frac{T_1}{T_2}$.

- (1) 1.5 (2) 1 (3) 0.5 (4) 2

Q32

An ideal gas undergoes a process where its pressure is directly proportional to its absolute temperature, and the temperature of the gas increases. Select the correct statements regarding this process.

- I. The volume of the gas remains constant.
 II. The work done by the gas is zero.
 III. The internal energy of the gas increases.
 IV. The heat absorbed by the gas is equal to the change in its internal energy.
 V. The process is an isothermal expansion.

- (1) I, II, III, IV Only (2) I, II, III Only
 (3) I, IV, V Only (4) II, III, IV Only

Q33

A charged particle, with mass M and charge q , begins its motion with an initial kinetic energy K_0 and moves along the positive x-axis. At time $t = 0$, it enters a region where a uniform electric field $\vec{E} = E\hat{j}$ is present. If the initial de Broglie wavelength of the particle is $\lambda_{initial}$, what will be its de Broglie wavelength after a duration τ ?

- (1) $\frac{\lambda_{initial}}{\sqrt{1 + \frac{q^2 E^2 \tau^2}{2MK_0}}}$ (2) $\lambda_{initial} \sqrt{1 + \frac{q^2 E^2 \tau^2}{2MK_0}}$
 (3) $\frac{\lambda_{initial}}{\sqrt{1 - \frac{q^2 E^2 \tau^2}{2MK_0}}}$ (4) $\lambda_{initial}$

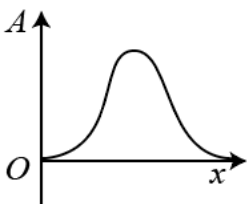
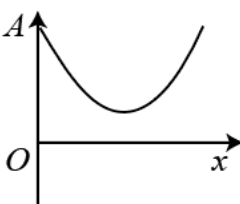
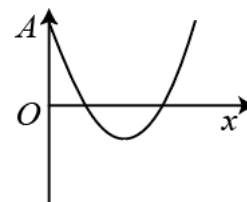
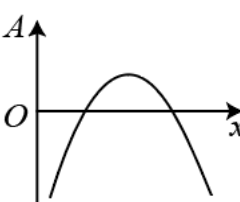
Q34

The distance between an object and a screen is 100 cm. A lens produces an image on the screen when placed at either of the two positions 40 cm apart. The power of the lens (in D) is nearly (mark answer to nearest integer)

- (1) 3 (2) 6 (3) 4 (4) 5

Q35

The potential energy U of a particle of mass $m = 3\text{kg}$ moving in the x -direction is given by $U = 3(x - 1) - (x - 3)^3$ When U is in joule and x is in metre. Which of the graphs shown in figure. Best represents the variation of the acceleration A of the particle with position x ?

- (1)  (2) 
 (3)  (4) 

Q36

For a prism, $A = 60^\circ$, refractive index $\mu = \sqrt{7/3}$. placed in air, the maximum possible angle of deviation, without suffering total internal reflection at the second surface is $\delta_{\max} = \frac{\pi}{n}$ (radian) find $n = ?$

- (1) 2 (2) 3 (3) 4 (4) 6

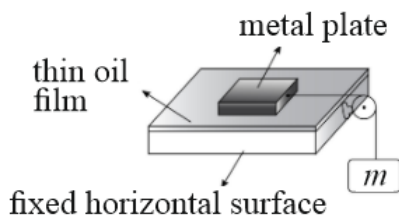
Q37

A small object undergoes simple harmonic motion with a period of 4 seconds and an amplitude of 3 cm. If it starts its oscillation from the equilibrium position, what is the ratio of the total distance traveled to the magnitude of its final displacement after 17 seconds?

- (1) 15 (2) 16 (3) 17 (4) 18

Q38

A rectangular metal plate has dimensions of $10\text{ cm} \times 20\text{ cm}$. A thin film of oil separates the plate from a fixed horizontal surface. The separation between the rectangular plate and the horizontal surface is 0.2 mm . An ideal string is attached to the plate and passes over an ideal pulley to a mass m . When $m = 125\text{ gm}$, the metal plate moves at a constant speed of 5 cm s^{-1} across the horizontal surface. Then, the coefficient of viscosity of oil in dyne s cm^{-2} is (Use $g = 1000\text{ cm s}^{-2}$)



- (1) 5 (2) 25 (3) 2.5 (4) 50

Q39

Two identical thin Plano-convex glass lenses (refractive index 1.5) each having radius of curvature of 20 cm are placed with their convex surfaces in contact at the center. The intervening space is filled with oil of refractive index 1.7. The focal length of the combination is

- (1) -20 cm (2) -25 cm (3) -50 cm (4) 50 cm

Q40

A small air pocket, with a radius of 0.2 cm , is trapped at a depth of 15 cm beneath the surface of a liquid. The liquid has a density of 900 kg/m^3 . If the absolute pressure inside this air pocket is measured to be 1500 N/m^2 higher than the ambient atmospheric pressure, what is the surface tension of the liquid in N/m ? (Assume acceleration due to gravity $g = 10\text{ m/s}^2$).

- (1) 0.10 (2) 0.15 (3) 0.20 (4) 0.25

Q41

A solid cylinder is rolling down on an inclined plane of angle θ . The coefficient of static friction between the plane and the cylinder is μ_s . The condition for the cylinder not to slip is

- (1) $\tan \theta \geq 3\mu_s$ (2) $\tan \theta > 3\mu_s$ (3) $\tan \theta \leq 3\mu_s$ (4) $\tan \theta < 3\mu_s$

Q42

In a Young's double-slit setup, two monochromatic light sources, one emitting light of wavelength 500 nm and the other 750 nm, are used simultaneously. If both lights produce their respective interference patterns on a screen, what is the lowest order bright fringe of the 500 nm light that perfectly overlaps with a bright fringe of the 750 nm light, excluding the central maximum?

- (1) 2 (2) 3 (3) 4 (4) 5

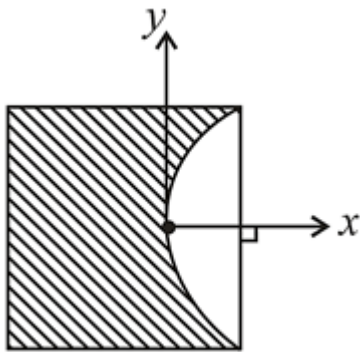
Q43

Consider a parallel plate capacitor where the plates each have an area A and are separated by a distance d . A uniform electric field of strength E is established between the plates. If the permittivity of the free space between the plates is ϵ_0 , what is the total energy stored in the electric field within the capacitor?

- (1) $\epsilon_0 E^2 A d$ (2) $\frac{1}{2} \epsilon_0 E^2 A d$ (3) $\frac{1}{4} \epsilon_0 E^2 A d$ (4) $\frac{3}{4} \epsilon_0 E^2 A d$

Q44

In the figure shown a semicircular area is removed from a uniform square plate of side ℓ and mass (before removing) m . The x -coordinate of centre of mass of remaining portion is (The origin is at the centre of square)



- (1) $-\frac{\pi(\pi-2)\ell}{2(8-\pi)}$ (2) $\frac{\pi(\pi-2)\ell}{2(8-\pi)}$ (3) $-\frac{\pi(\pi-2)\ell}{(8-\pi)}$ (4) None of these

Q45

A student measures the mass, radius, and height of a cylindrical block in a laboratory experiment. The recorded measurements with their respective uncertainties are:

Mass $m = (500 \pm 5)$ g

Radius $r = (2.0 \pm 0.1)$ cm

Height $h = (10.0 \pm 0.2)$ cm

What is the percentage error in the calculated density of the material of the cylindrical block?

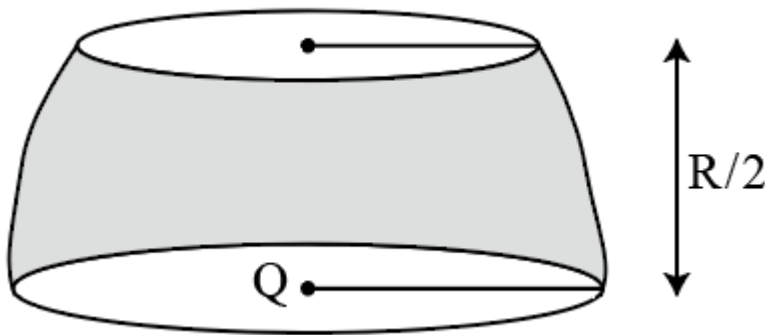
- (1) 10% (2) 13% (3) 15% (4) 12%

Q46

Two parallel wires in the plane of a paper are at a distance X_0 apart. A point charge is moving with a speed u between the wires in the same plane at a distance X_1 from one of the wires. When the wires carry current of magnitude I in the same direction, the radius of curvature of the path of the point charge is R_1 . In contrast, if the currents of magnitude I in the two wires have directions opposite to each other, the curvature of the path is R_2 . If $\frac{X_0}{X_1} = 3$, the value of $\frac{R_1}{R_2}$ is

Q47

A charge particle having charge Q is placed at the centre of hollow hemispherical sphere whose a portion is removed as shown. If the radius of the hollow sphere is R , then flux through the remaining surface is equal to $Q/x\epsilon_0$. The value of x is?



Q48

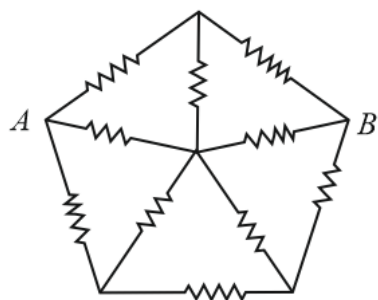
The specific heats of an ideal gas at constant pressure and constant volume are $525 \text{ J (kg}^\circ\text{C)}^{-1}$ and $315 \text{ J (kg}^\circ\text{C)}^{-1}$ respectively. Its density at NTP is (in $\frac{\text{Kg}}{\text{m}^3}$) (round off answer to nearest integer)

Q49

A screw gauge initially has a least count of 0.01 mm . If its pitch is doubled and the number of divisions on its circular scale is halved, what will be the new least count of the instrument? Express your answer in $\times 10^{-3} \text{ mm}$.

Q50

The effective resistance between A and B in the network shown, where the resistance of each resistor is R , is $16R/K$, find the integral value of K .



Q51

Which of the following statements accurately describe 2-deoxyribose, the carbohydrate component of DNA?

- I. It is a five-carbon monosaccharide.
- II. It predominantly forms a six-membered ring structure in its cyclic form.
- III. It belongs to the D-series of sugars.
- IV. When free in solution, it exhibits reducing properties.
- V. In the DNA polymer, it is linked via an α -glycosidic bond.

Choose the correct combination of statements:

- (1) I, III and IV Only
- (2) I, II and V Only
- (3) II, III and IV Only
- (4) I, IV and V Only

Q52

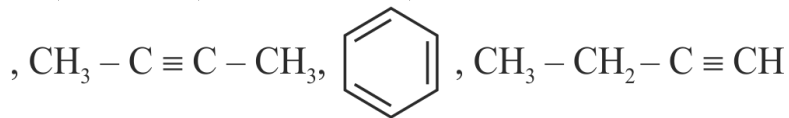
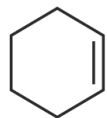
The correct order for increasing reactivity towards S_N1 substitution for the following compounds is

- 1. $\text{CH}_3\text{CH}_2\text{Br}$ 2. $\text{CH}_2 = \text{CH} - \text{CH}(\text{Br})\text{CH}_3$
- 3. $\text{CH}_2 = \text{CHBr}$ 4. $\text{CH}_3 - \text{CH}(\text{Br})\text{CH}_3$
- (1) $1 < 4 < 3 < 2$ (2) $4 < 1 < 2 < 3$
- (3) $3 < 1 < 2 < 4$ (4) $3 < 1 < 4 < 2$

Q53

How many of these compounds can react with ammonical silver nitrate solution to give white ppt.

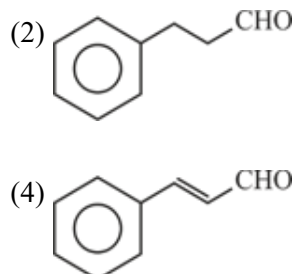
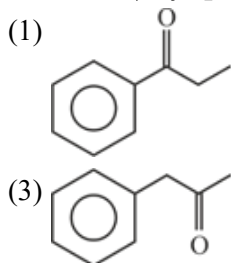
$\text{CH}_2 = \text{CH}_2$, $\text{CH} \equiv \text{CH}$, $\text{CH}_3 - \text{C} \equiv \text{CH}$, $\text{R} - \text{CH} = \text{CH}_2$



- (1) 2 (2) 3 (3) 4 (4) 5

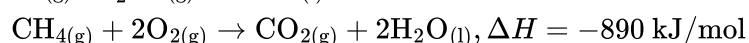
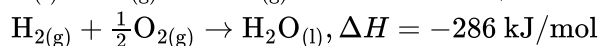
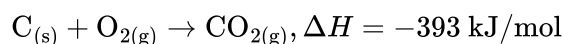
Q54

An organic compound having molecular formula $C_9H_{10}O$ is supplied to a student to determine the structural formula of compound, he observe that compound forms an orange red precipitate with 2,4 – *DNP* reagent and yellow precipitate on heating with $I_2/NaOH$. It neither reduces Tollen's reagent or Fehling solution nor it decolorises bromine water solution. On drastic oxidation with CrO_3 , it produces a carboxylic acid having molecular formula $C_7H_6O_2$. Identify the correct compound *A*

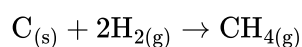


Q55

Consider the following thermochemical equations:



Calculate the standard enthalpy change for the formation of methane from its elements:



(1) -75 kJ/mol

(2) $+75 \text{ kJ/mol}$

(3) -150 kJ/mol

(4) $+150 \text{ kJ/mol}$

Q56

Chromyl chloride, a fuming red-brown liquid, reacts with an excess of aqueous sodium hydroxide. This reaction yields a yellow solution containing a sodium salt of chromic acid, denoted as compound 'A'. Subsequently, when an acidic solution of 'A' is treated with hydrogen peroxide, a characteristic deep blue coloration appears due to the formation of an unstable peroxo compound, 'B'. Identify the chemical formulas for 'A' and 'B'.

(1) $A = Na_2CrO_4, B = CrO_5$

(2) $A = Na_2Cr_2O_7, B = CrO_3$

(3) $A = Na_2CrO_4, B = Cr_2O_3$

(4) $A = Na_2Cr_2O_7, B = CrO_5$

Q57

Ammonium chloride solid is placed in an evacuated container and heated to 600 K. It decomposes according to the reaction: $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$. At equilibrium, the total pressure inside the container is measured to be 1.2 atm. Determine the value of the equilibrium constant K_p for this decomposition at 600 K.

(1) 0.36

(2) 0.6

(3) 1.2

(4) 1.44

Q58

2665mg of an octahedral complex $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ needs 3.4 g of AgNO_3 for complete precipitation of its all the free chloride ions. If the given complex is subjected to dehydration then the weight (in g) of anhydrous complex obtained will be x.

If your answer is x, report 2x after rounding off to nearest integer

- (1) 2 (2) 6 (3) 4 (4) 5

Q59

Identify the lanthanide ion from the given choices that is most readily oxidized to achieve its common and stable +3 oxidation state. (Atomic numbers: Nd = 60, Eu = 63, Dy = 66, Lu = 71)

- (1) Nd^{3+} (2) Eu^{2+} (3) Dy^{3+} (4) Lu^{3+}

Q60

The solubility product constants (K_{sp}) for Copper(II) sulfide (CuS), Zinc sulfide (ZnS), Calcium carbonate (CaCO_3), and Lead(II) chloride (PbCl_2) are provided below:

$$\text{CuS} : 8.5 \times 10^{-45}$$

$$\text{ZnS} : 2.5 \times 10^{-22}$$

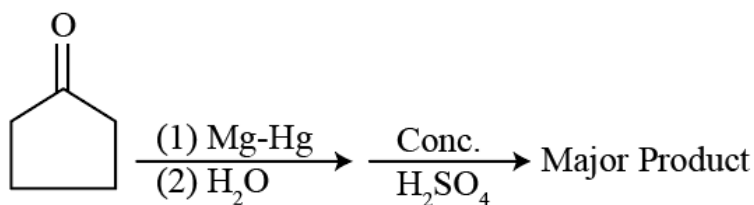
$$\text{CaCO}_3 : 3.4 \times 10^{-9}$$

$$\text{PbCl}_2 : 1.7 \times 10^{-5}$$

Arrange these compounds from the least soluble product to the most soluble product.

- (1) $\text{CuS} < \text{ZnS} < \text{CaCO}_3 < \text{PbCl}_2$ (2) $\text{PbCl}_2 < \text{CaCO}_3 < \text{ZnS} < \text{CuS}$
 (3) $\text{ZnS} < \text{CuS} < \text{PbCl}_2 < \text{CaCO}_3$ (4) $\text{CaCO}_3 < \text{PbCl}_2 < \text{CuS} < \text{ZnS}$

Q61



Major Product is:

- (1)
- (2)
- (3)
- (4)

Q62

Which of the following statements is correct about "shielding effect"?

- (1) It is effective, when the orbitals in the inner shells are completely filled.
- (2) Inner shells do not show any effect.
- (3) With increase of shielding down the group, ionisation energy increases.
- (4) With increasing nuclear charge, the shielding effect increases.

Q63

0.2 gm of an organic compound was analysed by Kjeldahl's method, Ammonia evolved was absorbed in 60 ml of $N/5$ H_2SO_4 . Unused acid required 40ml of $N/10$ $NaOH$ for complete neutralization. The percentage of nitrogen in the compound is :

- (1) 58
- (2) 66
- (3) 56
- (4) 68

Q64

Identify the correct statement(s) from the following

a) Dipole moment of NH_3 is more than NF_3 b) SF_4 is square planar c) $SnCl_4$ is more covalent than $SnCl_2$ d) I_2SO_4 sulphur atom has expanded octet

- (1) a,b
- (2) b,c,d
- (3) a,c,d
- (4) a,b,c

Q65

Assertion (A) : pK_a of phenol is 4.19 and that of benzoic acid is 10

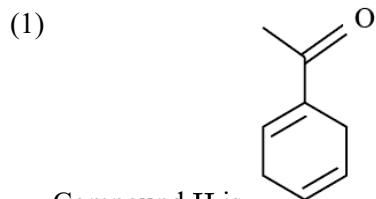
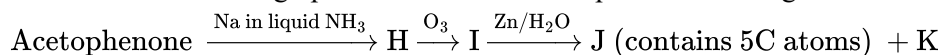
Reason (R) : Phenoxide ion is stabilised by nonequivalent resonance structures whereas benzoate ion by two equivalent resonance structures

The correct option among the following is

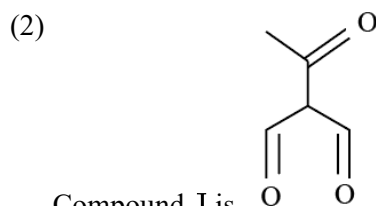
- (1) A and R are true. R is the correct explanation of A
- (2) A and R are true, but R is not the correct explanation for A
- (3) A is true R is false
- (4) A is false but R is true

Q66

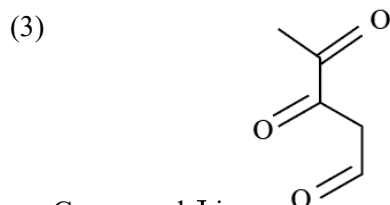
Which of the following options is correct for the products of the given reaction sequence?



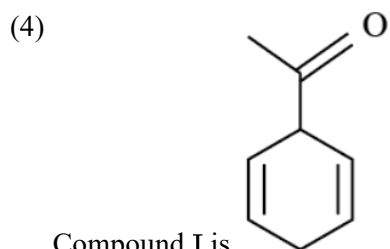
Compound H is



Compound J is



Compound J is



Compound I is

Q67

The limiting molar conductivities (λ°) for a selection of cations in water at 25°C are given as:

$$\lambda^\circ(\text{H}^+) = 349.8 \text{ Scm}^2 \text{ mol}^{-1}$$

$$\lambda^\circ(\text{Na}^+) = 50.11 \text{ Scm}^2 \text{ mol}^{-1}$$

$$\lambda^\circ(\text{K}^+) = 73.52 \text{ Scm}^2 \text{ mol}^{-1}$$

$$\lambda^\circ(\text{Ca}^{2+}) = 119 \text{ Scm}^2 \text{ mol}^{-1}$$

$$\lambda^\circ(\text{Mg}^{2+}) = 106.12 \text{ Scm}^2 \text{ mol}^{-1}$$

Which of the following statements regarding the relative limiting molar conductivities of these ions is INCORRECT?

- (1) The limiting molar conductivity of H^+ is significantly higher than that of any other cation listed.
- (2) Ca^{2+} has a higher limiting molar conductivity than Mg^{2+} .
- (3) K^+ exhibits a lower limiting molar conductivity compared to Na^+ .
- (4) Among the monovalent alkali metal ions listed, K^+ has a higher limiting molar conductivity than Na^+ .

Q68

The elements having the maximum and minimum melting points among the members of the group 13 respectively are

- (1) B, Tl (2) Tl, B (3) B, Ga (4) Al, Tl

Q69

2 g of a non-electrolyte solute (molar mass is 500 g mol^{-1}) was dissolved in 57.3 g of xylene. If the freezing point depression constant K_f of xylene is $4.3 \text{ K kg mol}^{-1}$. Then, the depression in freezing point of xylene is.....

- (1) 57.3 K (2) 0.3 K (3) 4.3 K (4) 0.002 K

Q70

Consider the following statements regarding the properties of ammonia (NH_3) and phosphine (PH_3):

Statement I: Ammonia (NH_3) has a higher boiling point than phosphine (PH_3).

Statement II: Nitrogen, being more electronegative and smaller than phosphorus, enables NH_3 molecules to form intermolecular hydrogen bonds, which are significantly stronger than the Van der Waals forces present between PH_3 molecules.

Choose the correct option:

- (1) Both Statement I and Statement II are correct, and Statement II is the correct explanation for Statement I.
(2) Both Statement I and Statement II are correct, but Statement II is not the correct explanation for Statement I.
(3) Statement I is correct, but Statement II is incorrect.
(4) Statement I is incorrect, but Statement II is correct.

Q71

A laboratory experiment involves the reaction of benzoic acid with an aqueous solution of sodium bicarbonate. If the reaction yields 11.2 liters of carbon dioxide gas when measured at standard temperature and pressure (STP), calculate the mass (in grams) of sodium bicarbonate that participated in this reaction. (Given: Molar mass of $\text{NaHCO}_3 = 84 \text{ g/mol}$, Molar volume of gas at STP = 22.4 L/mol).

Q72

For the chemical reaction $2P(g) + Q(g) \rightleftharpoons P_2Q(g)$, the standard molar entropies are given as

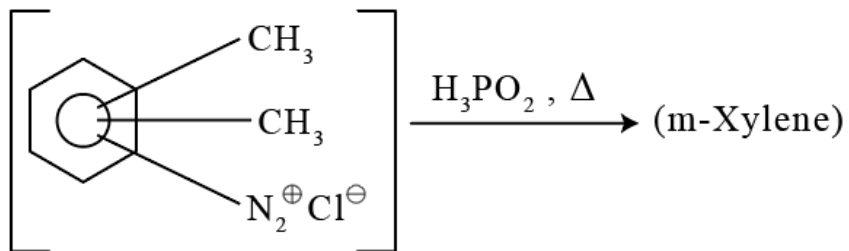
$S^\ominus(P) = 40 \text{ J K}^{-1} \text{ mol}^{-1}$, $S^\ominus(Q) = 90 \text{ J K}^{-1} \text{ mol}^{-1}$, and $S^\ominus(P_2Q) = 120 \text{ J K}^{-1} \text{ mol}^{-1}$. If the standard enthalpy change for the reaction is $\Delta H^\ominus = -25 \text{ kJ mol}^{-1}$, calculate the temperature in Kelvin at which the reaction reaches equilibrium. (Nearest integer)

Q73

An acid 'A' is 0.4 N. In a neutralization process 50ml of 'A' is mixed with 30ml of sodium carbonate solution (42.4 g/Litre) and 10 mL of the resulting solution is diluted to 50 mL and then 5 mL of this solution (B) is completely neutralised with certain volume of 40 times diluted solution of (A). find the volume of diluted solution of (A) required for the neutralization of B (in mL)

Q74

How many aromatic diazonium chlorides of xylene can be reduced to meta-xylene by H_3PO_2 ?



Q75

Consider a series of aqueous solutions, each containing one of the following metal cations:

Ni^{2+} , Co^{2+} , Mn^{2+} , Al^{3+} , Cr^{3+} , Ag^+ , Pb^{2+} . When a solution of potassium ferrocyanide, $\text{K}_4[\text{Fe}(\text{CN})_6]$, is added to each of these solutions, how many of these cations will produce a characteristic precipitate?

ANSWERS AND SOLUTIONS

1. (3)	2. (4)	3. (4)	4. (2)	5. (4)	6. (2)	7. (3)	8. (2)
9. (3)	10. (2)	11. (2)	12. (1)	13. (2)	14. (1)	15. (1)	16. (4)
17. (3)	18. (2)	19. (4)	20. (2)	21. 216.0	22. 2.0	23. 72.0	24. 11.0
25. 2.0	26. (2)	27. (4)	28. (4)	29. (2)	30. (4)	31. (2)	32. (1)
33. (1)	34. (4)	35. (3)	36. (2)	37. (3)	38. (3)	39. (3)	40. (2)
41. (3)	42. (2)	43. (2)	44. (4)	45. (2)	46. 3.0	47. 4.0	48. 2.0
49. 40.0	50. 22.0	51. (1)	52. (4)	53. (2)	54. (3)	55. (1)	56. (1)
57. (1)	58. (4)	59. (2)	60. (1)	61. (4)	62. (1)	63. (3)	64. (3)
65. (4)	66. (2)	67. (3)	68. (3)	69. (2)	70. (1)	71. 42.0	72. 500.0
73. 5.0	74. 3.0	75. 5.0					